

AI QUIZ GENERATOR

Multi-Step Generative AI Workflow for Automatic MCQ Generation, Evaluation & Explanations

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ASSESSMENT	Mini Assessment — GenAI Workflow Design

Assessment Alignment

This assessment applies the required three-stage pattern from the assessment-rules document:

Structuring & Planning → **Drafting/Execution** → **Critique & Refinement**. The final stage acts as a mandatory evaluation gate before the work is finalized.

Project Context

Concept: The AI Quiz Generator is a prompt-based Generative AI application that creates multiple-choice questions from a user-selected topic. It generates four options, identifies the correct answer, evaluates the user's responses, calculates a score, and provides concise explanations.

Objective: Design a prompt-based GenAI workflow that accepts topic, question count, and difficulty; generates MCQs; evaluates answers; calculates the score; and provides explanations and revision suggestions.

Source workflow used

The rules require a clear progression, dependency between steps, and a mandatory self-evaluation/refinement gate. The assessment below follows that structure for the AI Quiz Generator.

STEP 1 — STRUCTURING & PLANNING (THE SETUP)

Goal: Establish the AI persona, provide the initial project information, and organize the requirements into a structured plan before generating the final assessment content.

Prompt 1 — Requirement Extraction & Workflow Outline

Act as an expert Generative AI workflow designer and educational quiz specialist. I am going to give you the project information for an “AI Quiz Generator” that creates MCQs from a topic entered by a user.

Your task is to extract the main functional requirements and create a structured workflow outline for the assessment. Include: user input, question generation, four answer options, one correct answer, answer evaluation, score calculation, explanations, weak-area identification, and revision suggestions.

Keep the workflow logically ordered and make each stage depend on the output of the previous stage. Use the project information provided below and do not introduce unrelated features.

[Insert AI Quiz Generator project information here]

Expected AI Output

- A structured outline identifying the six main stages of the workflow.
- Clear inputs and outputs for each stage.
- A dependency chain from topic and preferences through generated MCQs, evaluation, feedback, and revision.

Structured Outline

Stage	Input	Output
1. Understand User Input	Topic + difficulty + question count	Quiz requirements
2. Generate Questions	Quiz requirements	Generated MCQs
3. Generate Answer Key	MCQs	Correct answer + plausible distractors
4. Evaluate Answers	User responses + answer key	Score + question results
5. Generate Feedback	Question results	Explanations + learning feedback
6. Suggest Revision	Score + weak areas	Suggested revision topics

STEP 2 — DRAFTING (THE EXECUTION)

Goal: Generate the assessment output strictly from the structured workflow produced in Step 1.

Prompt 2 — Generate the AI Quiz Workflow

Excellent. Now, using only the structured outline you just generated, create the first complete draft of the AI Quiz Generator workflow.

Constraints:

- Accept topic, number of questions, and difficulty level as dynamic user inputs.
- Generate clear MCQs with exactly four options and one correct answer.
- Keep the generated questions relevant to the selected topic and difficulty.
- Evaluate submitted answers against the generated answer key.
- Calculate the total score and show question-level results.
- Generate a concise explanation for each answer.
- Identify weak areas and suggest revision topics.
- Keep the workflow sequential and show the output of each stage.
- Use prompt chaining and structured output terminology where appropriate.
- Do not add features that are not supported by the project information.

Expected AI Output: First Draft

- Stage 1 — Input:** Collect topic, number of questions, and difficulty level.
- Stage 2 — Generation:** Generate relevant MCQs with four options and one correct answer.
- Stage 3 — Answer Key:** Store the correct answer and plausible distractors in a structured format.
- Stage 4 — Attempt & Evaluation:** Present the quiz, compare user responses with the answer key, and calculate the score.
- Stage 5 — Explanation:** Explain correct answers and clarify mistakes.
- Stage 6 — Revision:** Use score and weak areas to suggest revision topics.

Demonstration Input

Topic: Machine Learning **Difficulty:** Medium **Questions:** 5

Demonstration Output Flow

INPUT	GENERATION	USER ACTION	EVALUATION	FINAL OUTPUT
Machine Learning + Medium	5 MCQs + 4 options + answer key	Select one answer per question	Compare responses with answer key	Score, correct answers + explanations + revision topics

STEP 3 — CRITIQUE & REFINEMENT (THE REQUIRED GATE)

Goal: Force the AI to evaluate the draft against explicit criteria and correct any failures before finalization.

Prompt 3 — Strict Senior Reviewer

Before we finalize this assessment, act as a strict senior reviewer for Generative AI workflow design. Review the draft you just produced against the following checklist:

1. Does the workflow clearly begin with topic, question count, and difficulty as user inputs?
2. Does question generation produce four options and exactly one correct answer?
3. Is there a clear dependency between each stage of the prompt chain?
4. Does evaluation compare user responses with the answer key and calculate a score?
5. Does the workflow provide explanations for correct and incorrect answers?
6. Does the final stage identify weak areas and suggest revision topics?
7. Is structured output and prompt chaining represented consistently?
8. Are all stages relevant to the stated AI Quiz Generator project?

First, list your critique and identify any failed checklist items. Second, output the final corrected workflow that fixes those issues. Do not remove required functionality.

Self-Critique

- **Checklist 1 — Passed:** The workflow starts with topic, question count, and difficulty as user inputs.
- **Checklist 2 — Passed:** The generation stage specifies four options and one correct answer.
- **Checklist 3 — Passed:** The six-stage chain creates a dependency from requirements to revision suggestions.
- **Checklist 4 — Passed:** User responses are compared with the answer key and converted into a score and question results.
- **Checklist 5 — Passed:** The feedback stage provides concise explanations and clarifies mistakes.
- **Checklist 6 — Passed:** Score and weak areas are used to produce suggested revision topics.
- **Checklist 7 — Passed:** Prompt chaining and structured output are explicitly included.
- **Checklist 8 — Passed:** The workflow remains focused on automatic MCQ generation, evaluation, explanations, and revision support.

Final Corrected Workflow

1. **Topic & Preference Input** → Collect topic, number of questions, and difficulty level.

2. Question Generation → Generate clear, topic-relevant MCQs with exactly four options and one correct answer.

3. Answer Key & Structured Output → Store each question, options, correct answer, and explanation in a predictable structure.

4. Quiz Attempt & Evaluation → Present questions to the user, compare selected answers with the answer key, and calculate the total score.

5. Explanation & Feedback → Explain why answers are correct or incorrect and identify mistakes.

6. Revision Suggestions → Use the score and weak areas to recommend topics for further revision.

Final Evaluation Gate

The workflow is accepted only after the reviewer confirms that all required inputs, outputs, dependencies, evaluation criteria, explanations, and revision suggestions are present. This satisfies the mandatory critique-and-refinement gate.

FINAL ASSESSMENT SUMMARY

The completed AI Quiz Generator demonstrates a multi-step Generative AI workflow rather than a single prompt. It connects user input, question generation, answer-key creation, quiz evaluation, feedback, and revision through dependent stages.

GenAI Components

- **Prompt Templates:** Keep question-generation instructions consistent while inserting dynamic topic and difficulty values.
- **LLM:** Generates questions, options, answer keys, explanations, and revision suggestions.
- **Structured Output:** Stores question, options, correct answer, and explanation in a predictable format.
- **Prompt Chaining:** Connects generation, answer evaluation, feedback, and revision stages.
- **Prompt Evaluation:** Tests relevance, correctness, uniqueness, and difficulty.

Key Benefits

- Automatically creates practice questions from a selected topic.
- Supports different difficulty levels.
- Saves time compared with manually creating quizzes.
- Provides instant scoring and explanations.
- Helps identify weak areas for revision.
- Demonstrates Generative AI, prompt engineering, structured output, and prompt chaining.

Suggested Technology Stack

Layer	Technology / Option
Language	Python
LLM	OpenAI / Gemini / other course-approved LLM
Prompt Framework	LangChain or direct API integration
Interface	Streamlit
Data Storage	JSON / SQLite (optional)
Output	Interactive quiz + score + explanations
Evaluation	Fixed quiz scenarios + quality rubric

Assessment conclusion: The final workflow follows the required progression of setup, execution, and mandatory self-review/refinement, with clear dependency between stages.